Berita Acara Perkuliahan

Mata Kuliah : 16TNK4053 Teknik Telekomunikasi (Sistem Komunikasi 1)

Kelas : 2NK

Jumlah Mahasiswa : 29 Orang

Minggu Ke- : 2

Waktu Kuliah : Senin, 07 Februari 2022

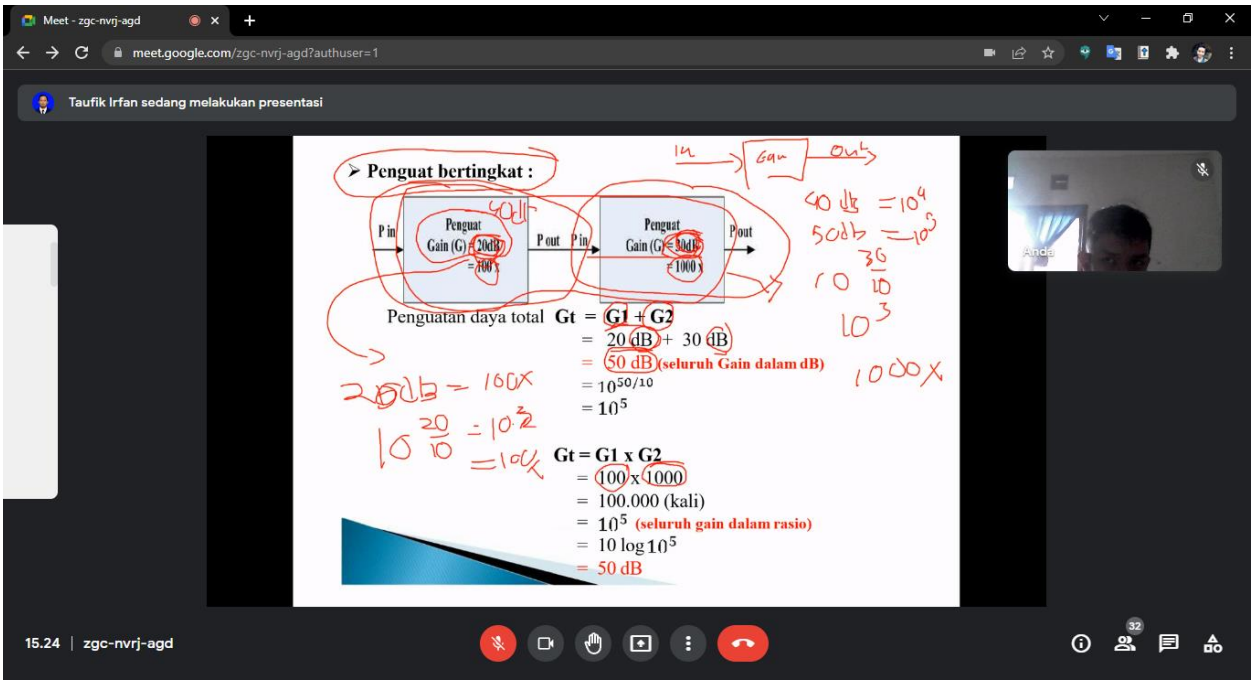
Materi :

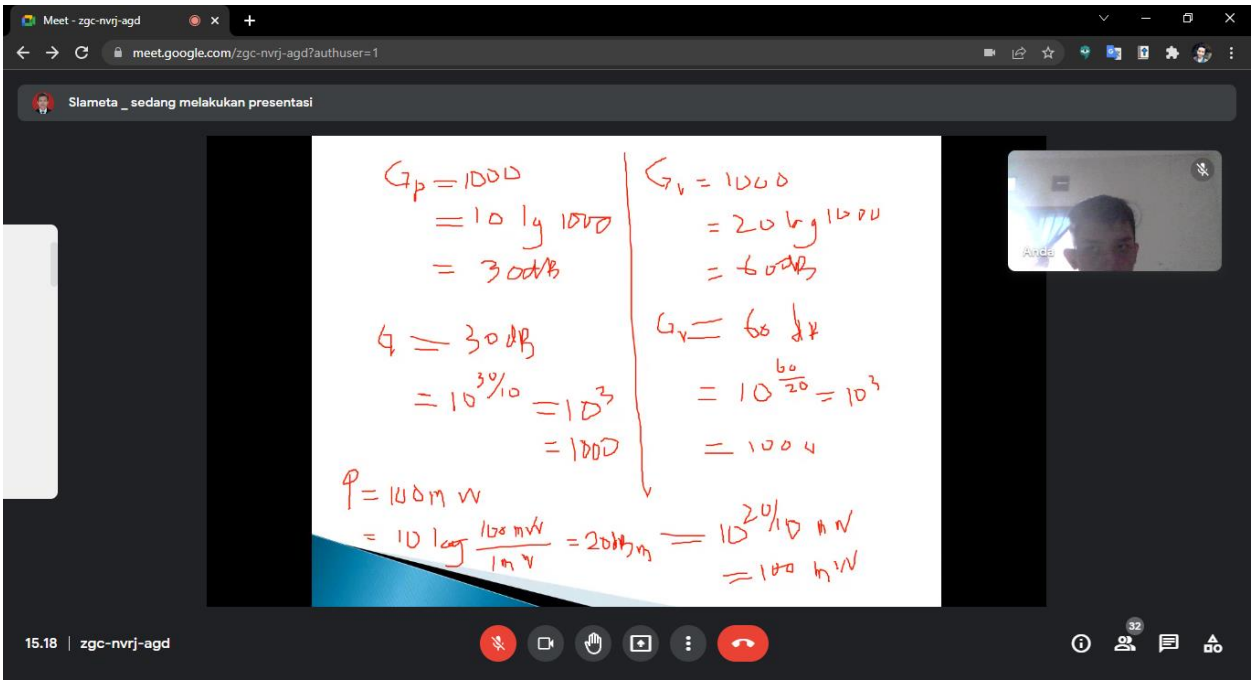
1. Internal Noise
2. Noise Figure
3. Modulasi dan Demodulasi

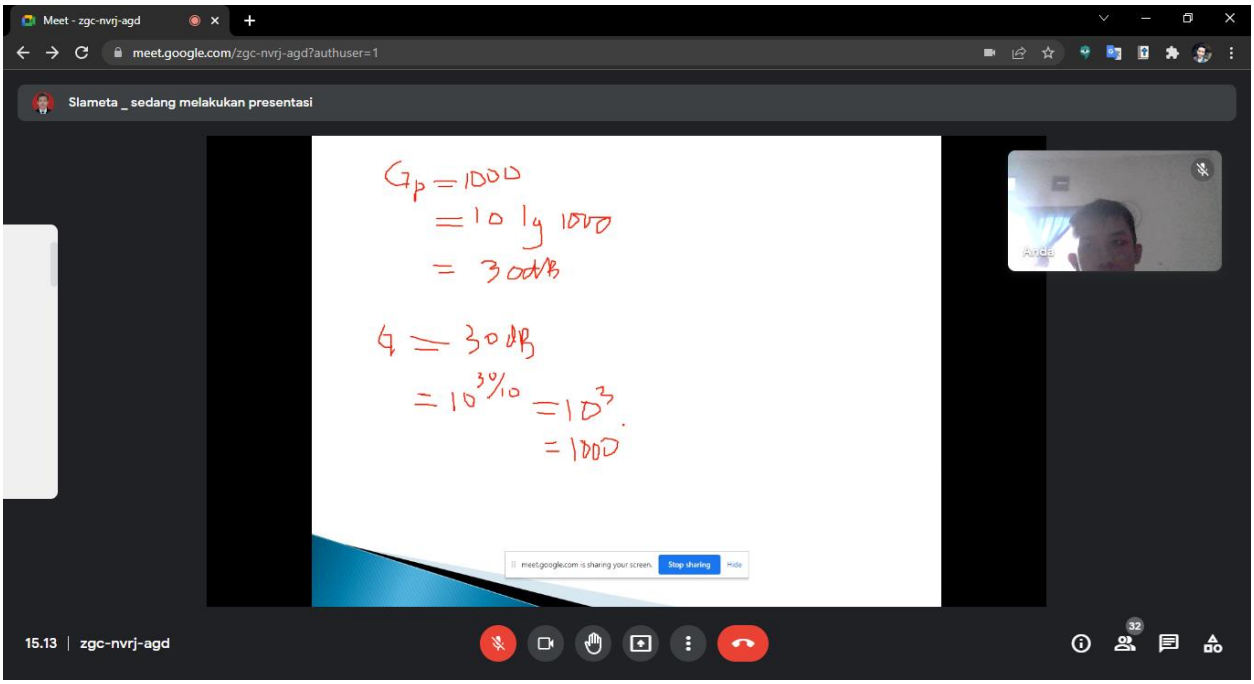
Kegiatan Mahasiswa :

1. Mendengarkan
2. Sesi tanya jawab

Bukti Pelaksanaan Kuliah (menggunakan Google Meeting)



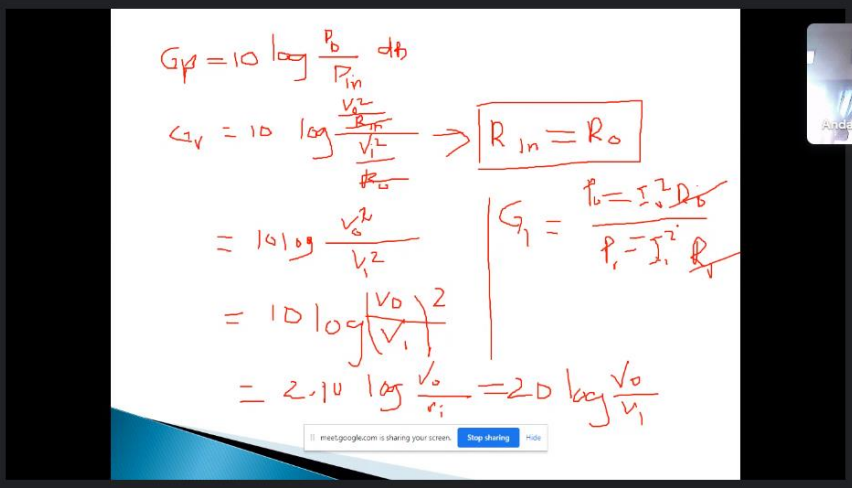

Meet - zgc-nvrj-agd
meet.google.com/zgc-nvrj-agd?authuser=1
Slameta _ sedang melakukan presentasi
$G_p = 1000$
$= 10 \lg 1000$
$= 30 dB$
$G = 30 dB$
$= 10^{30/10} = 10^3$
$= 1000$
$G_v = 1000$
$= 20 \lg 1000$
$= 60 dB$
$G_v = 60 dB$
$= 10^{\frac{60}{20}} = 10^3$
$= 1000$
$P = 100 mW$
$= 10 \lg \frac{100 mW}{1 mW} = 20 dB_m = 10^{20/10} mV$
$= 100 mV$
Anda
15.18 | zgc-nvrj-agd


Meet - zgc-nvrj-agd
meet.google.com/zgc-nvrj-agd?authuser=1
Slameta _ sedang melakukan presentasi
$G_p = 1000$
$= 10 \lg 1000$
$= 30 dB$
$G = 30 dB$
$= 10^{30/10} = 10^3$
$= 1000$
meet.google.com is sharing your screen.
Stop sharing
Hide
Anda
15.13 | zgc-nvrj-agd

Meet - zgc-nvrj-agd

meet.google.com/zgc-nvrj-agd?authuser=1

Slameta _ sedang melakukan presentasi





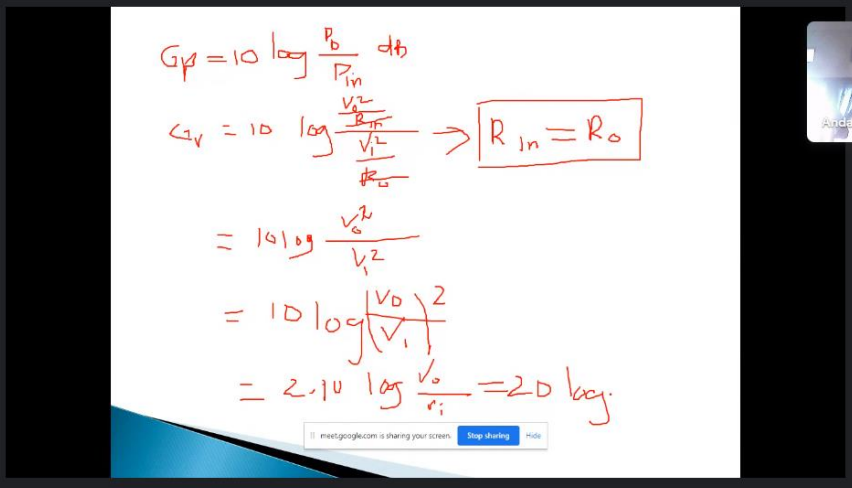
15.09 | zgc-nvrj-agd

32

Meet - zgc-nvrj-agd

meet.google.com/zgc-nvrj-agd?authuser=1

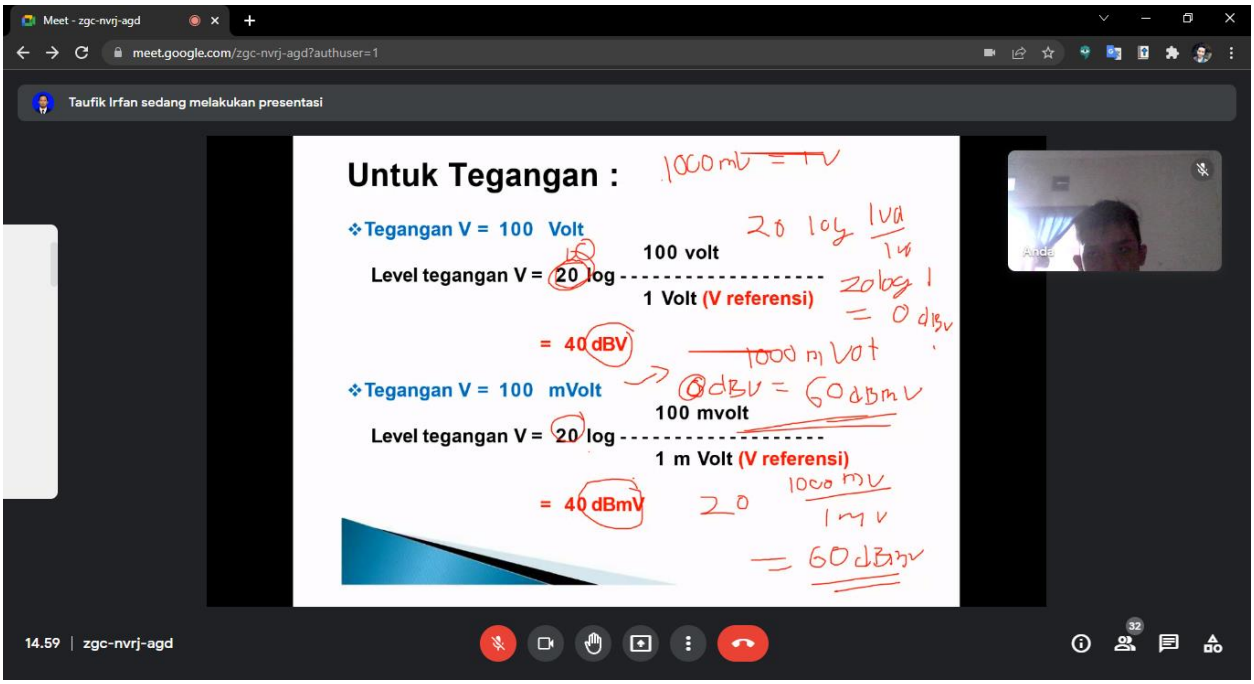
Slameta _ sedang melakukan presentasi

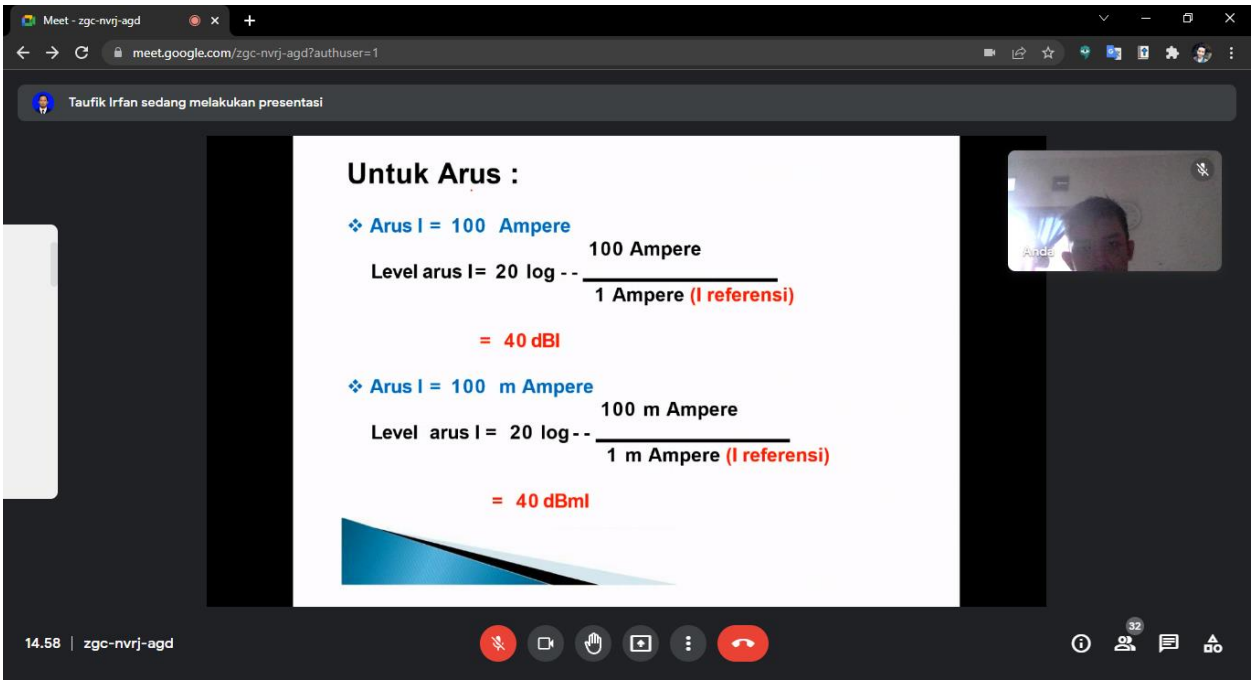


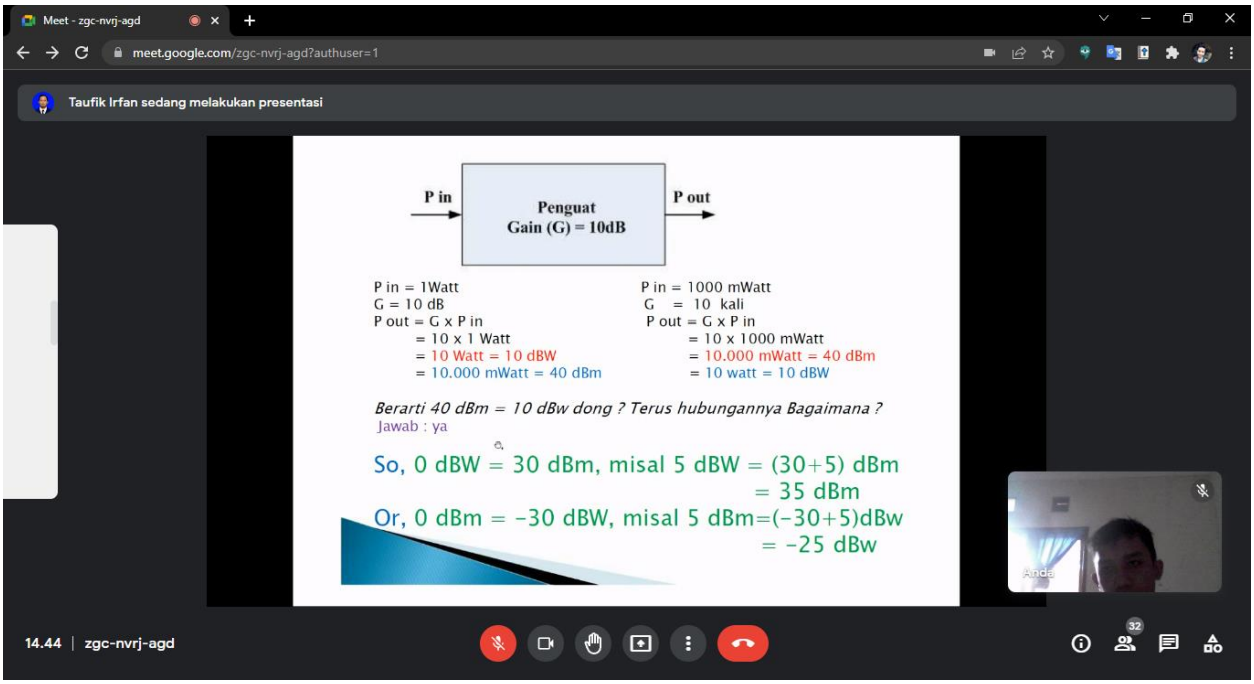


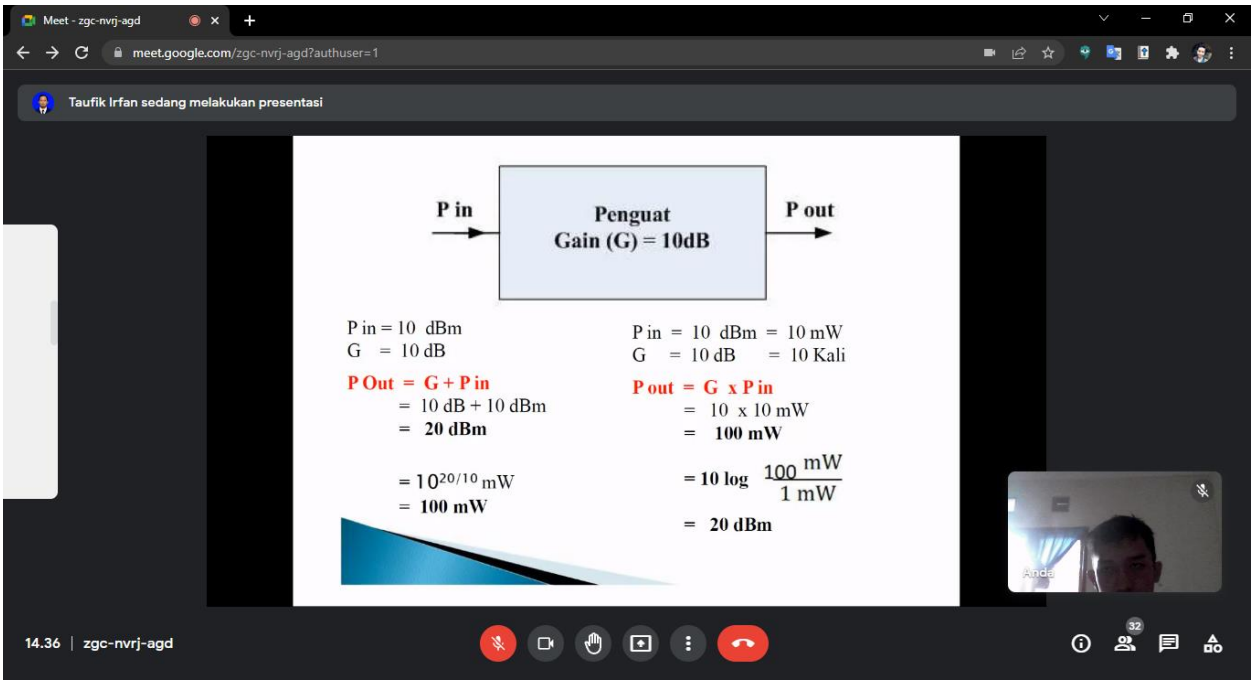
15.08 | zgc-nvrj-agd

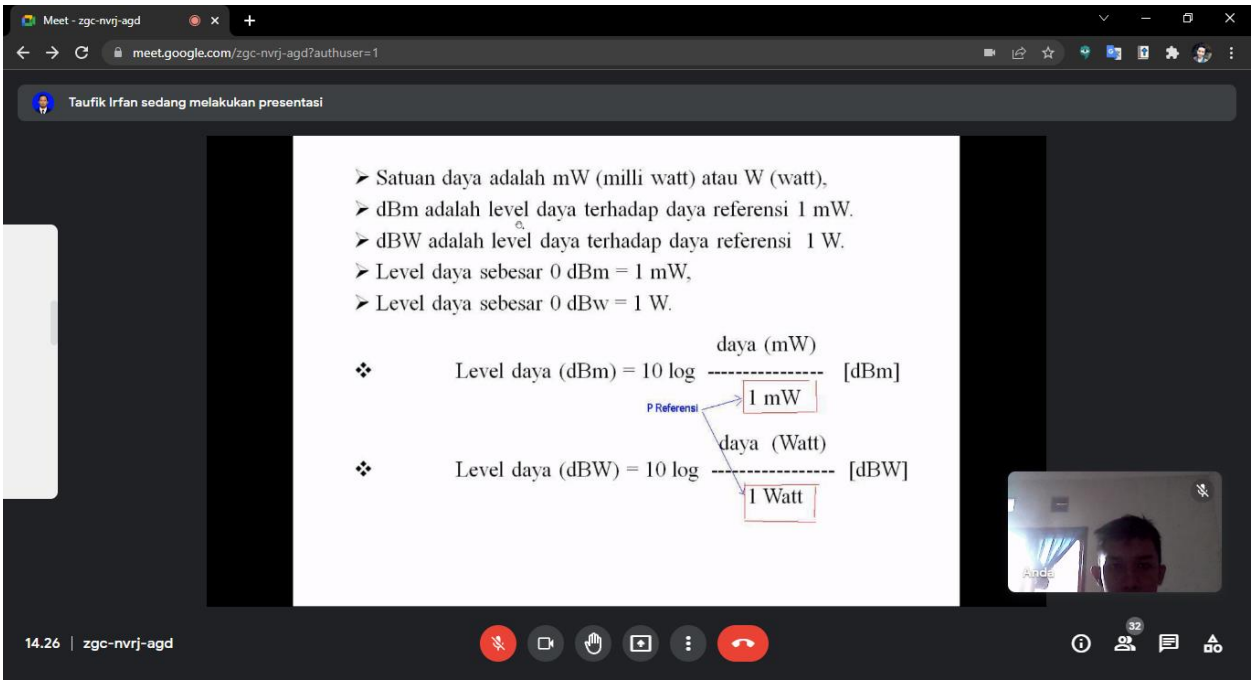
32

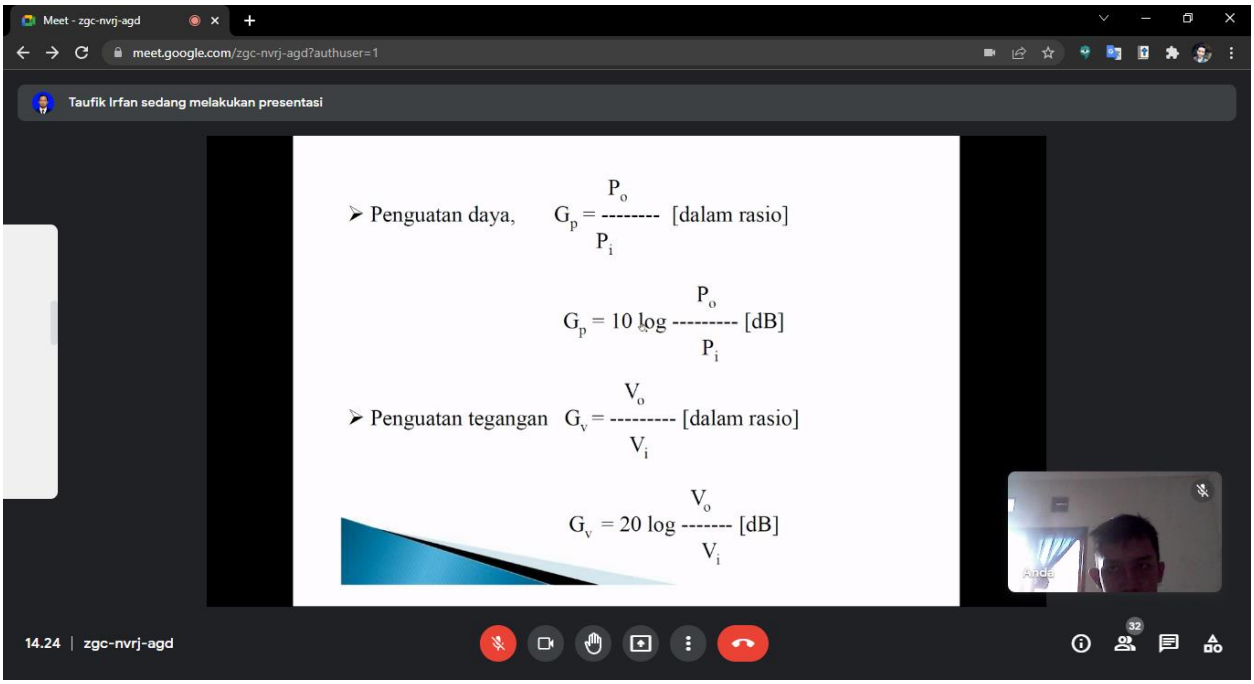

Meet - zgc-nvrj-agd
meet.google.com/zgc-nvrj-agd?authuser=1
Taufik Irfan sedang melakukan presentasi
Untuk Tegangan :
1000 mV = 1 V
❖ Tegangan V = 100 Volt
Level tegangan V = 20 log 100 volt / 1 Volt (V referensi)
20 log 1V/1V
20 log 1 = 0 dBV
= 40 dBV
1000 mVolt
0 dBV = 60 dBmV
❖ Tegangan V = 100 mVolt
Level tegangan V = 20 log 100 mvolt / 1 m Volt (V referensi)
= 40 dBmV
20 1000 mV / 1 mV
= 60 dBmV
14.59 | zgc-nvrj-agd

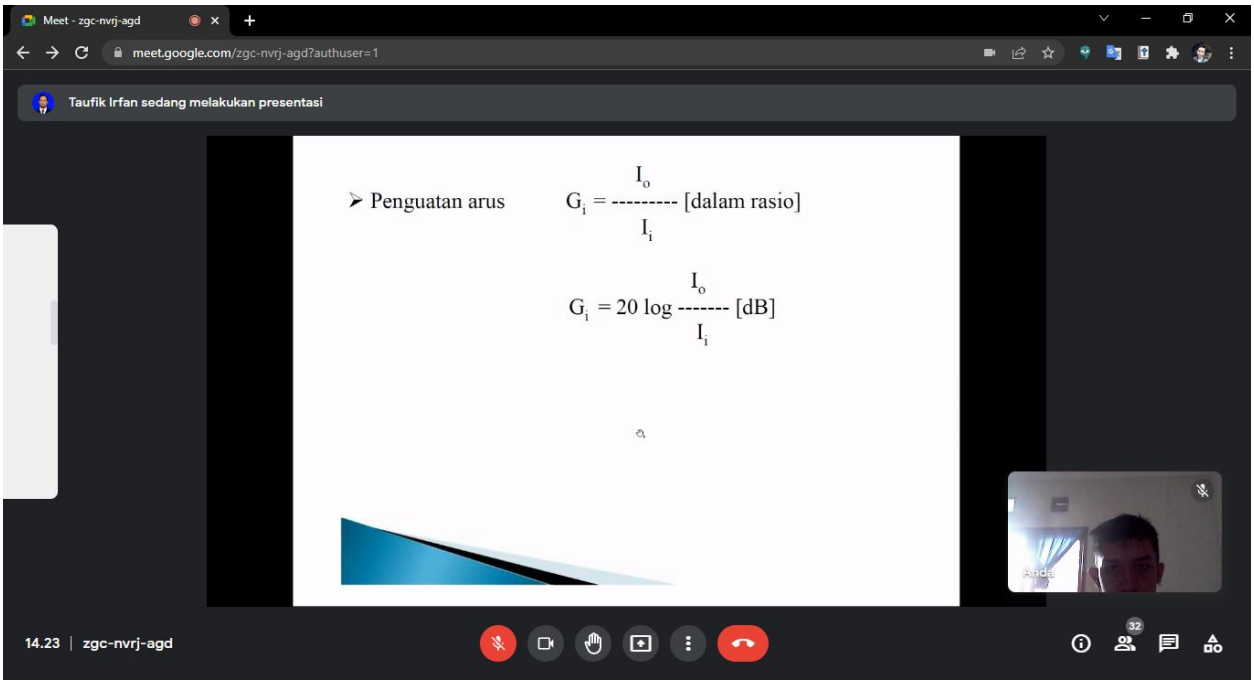

Meet - zgc-nvrj-agd
meet.google.com/zgc-nvrj-agd?authuser=1
Taufik Irfan sedang melakukan presentasi
Untuk Arus :
❖ Arus I = 100 Ampere
Level arus I = 20 log 100 Ampere / 1 Ampere (I referensi)
= 40 dBI
❖ Arus I = 100 m Ampere
Level arus I = 20 log 100 m Ampere / 1 m Ampere (I referensi)
= 40 dBmI
14.58 | zgc-nvrj-agd

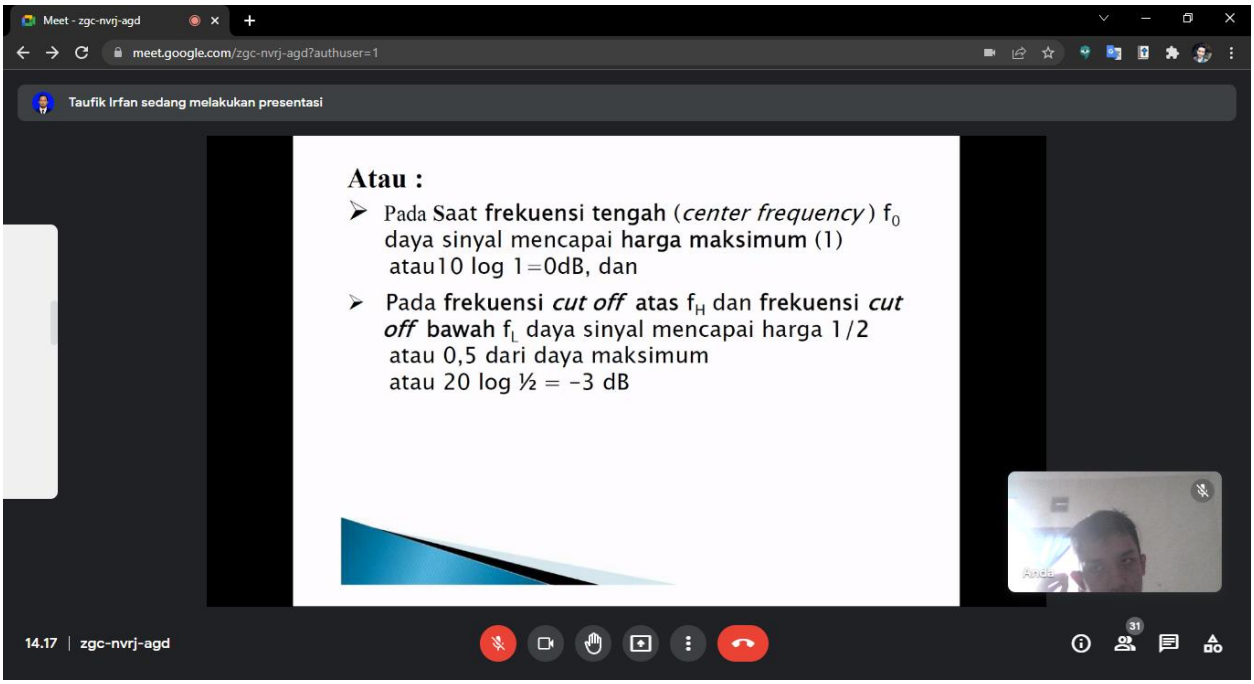

Meet - zgc-nvrj-agd
meet.google.com/zgc-nvrj-agd?authuser=1
Taufik Irfan sedang melakukan presentasi
P in
Penguat
Gain (G) = 10dB
P out
P in = 1Watt
G = 10 dB
P out = G x P in
= 10 x 1 Watt
= 10 Watt = 10 dBW
= 10.000 mWatt = 40 dBm
P in = 1000 mWatt
G = 10 kali
P out = G x P in
= 10 x 1000 mWatt
= 10.000 mWatt = 40 dBm
= 10 watt = 10 dBW
Berarti 40 dBm = 10 dBw dong ? Terus hubungannya Bagaimana ?
Jawab : ya
So, 0 dBW = 30 dBm, misal 5 dBW = (30+5) dBm
= 35 dBm
Or, 0 dBm = -30 dBW, misal 5 dBm=(-30+5)dBw
= -25 dBw
14.44 | zgc-nvrj-agd


Meet - zgc-nvrj-agd
meet.google.com/zgc-nvrj-agd?authuser=1
Taufik Irfan sedang melakukan presentasi
P in
Penguat
Gain (G) = 10dB
P out
P in = 10 dBm
G = 10 dB
P Out = G + P in
= 10 dB + 10 dBm
= 20 dBm
= 10^(20/10) mW
= 100 mW
P in = 10 dBm = 10 mW
G = 10 dB = 10 Kali
P out = G x P in
= 10 x 10 mW
= 100 mW
= 10 log 100 mW / 1 mW
= 20 dBm
14.36 | zgc-nvrj-agd


Meet - zgc-nvrj-agd
meet.google.com/zgc-nvrj-agd?authuser=1
Taufik Irfan sedang melakukan presentasi
➢ Satuan daya adalah mW (milli watt) atau W (watt),
➢ dBm adalah level daya terhadap daya referensi 1 mW.
➢ dBW adalah level daya terhadap daya referensi 1 W.
➢ Level daya sebesar 0 dBm = 1 mW,
➢ Level daya sebesar 0 dBw = 1 W.
❖ Level daya (dBm) = 10 log daya (mW) / 1 mW [dBm]
P Referensi
❖ Level daya (dBW) = 10 log daya (Watt) / 1 Watt [dBW]
14.26 | zgc-nvrj-agd


Meet - zgc-nvrj-agd
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Taufik Irfan sedang melakukan presentasi
➢ Penguatan daya, $G_p = \frac{P_o}{P_i}$ [dalam rasio]
$G_p = 10 \log \frac{P_o}{P_i}$ [dB]
➢ Penguatan tegangan $G_v = \frac{V_o}{V_i}$ [dalam rasio]
$G_v = 20 \log \frac{V_o}{V_i}$ [dB]
14.24 | zgc-nvrj-agd


Meet - zgc-nvrj-agd
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Taufik Irfan sedang melakukan presentasi
➢ Penguatan arus $G_i = \frac{I_o}{I_i}$ [dalam rasio]
$G_i = 20 \log \frac{I_o}{I_i}$ [dB]
14.23 | zgc-nvrj-agd


Meet - zgc-nvrj-agd
meet.google.com/zgc-nvrj-agd?authuser=1
Taufik Irfan sedang melakukan presentasi
Atau :
➢ Pada Saat frekuensi tengah (*center frequency*) f_0 daya sinyal mencapai harga maksimum (1) atau10 log 1=0dB, dan
➢ Pada frekuensi *cut off* atas f_H dan frekuensi *cut off* bawah f_L daya sinyal mencapai harga 1/2 atau 0,5 dari daya maksimum atau 20 log ½ = -3 dB
14.17 | zgc-nvrj-agd


Slameta
Muhammad Akmal Aman...
Michael Galileo Hasiari
Nabila Alia Zahra
MEITHA NOER MEIDAYAN...
Iqbal Dzhurfan
Nifa Syahla
Farhan Mulya Wijaya
Bagas Nurhidayat
Ihsan Anugerah
AXEL EGA ADIPUTRA
M Fadlan Azhary
Siti Nur Salamah
FIRDAUS SURYA PRATAMA
Iswatun Hasanah
Liza Febbyanti
Vanya Annisa Maharani
Ardi Anggrah
Nazma Nurfaidah Shiyami
Anda
zgc-nvrj-agd


Slameta
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Nabila Alia Zahra
MEITHA NOER MEIDAYAN...
Iqbal Dzhurfan
Nifa Syahla
Farhan Mulya Wijaya
Bagas Nurhidayat
Ihsan Anugerah
AXEL EGA ADIPUTRA
M Fadlan Azhary
Siti Nur Salamah
FIRDAUS SURYA PRATAMA
Iswatun Hasanah
Liza Febbyanti
Vanya Annisa Maharani
Ardi Anggrah
Nazma Nurfaidah Shiyami
Anda
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